# How To Check FOOD ADULTERATION

**Dr S.S. Verma** is professor at Department of Physics, Sant Longowal Institute of Engineering and Technology, Sangrur, Punjab

Due to changing social values where companies are focussing on reaping rich profits, adulteration is prevalent in almost all commodities such as food items, fuel, cloths and construction materials. Adulteration has reached alarming levels and is, hence, making detrimental effects to the well-being of humans.

Innocent lives are lost when infrastructures like flyovers, bridges, buildings and roads face unexpected collapses due to the use of sub-standard construction materials. The biggest paradox is that people who are involved in adulteration never think that they or their near and dear ones could also be the victims of this menace.

***Nowadays, adulteration is being done in sophisticated ways that demand cutting-edge research for the detection of adulterants***

## Food adulteration

Adulteration of food not only involves intentional addition or substitution of specific substances, but also biological and chemical contamination. The most common form of food adulteration is the addition of sub-standard substances (adulterants) to increase the quantity of raw or prepared food items.

Adulteration can happen at any level, from production, processing, storage to transportation and distribution. People involved are not concerned about the negative effects adulteration can have on the health of consumers. This article focuses on the development, use and benefits of electronic and optoelectronic devices to check food adulteration.

## Electronic nose

Human beings do not have the foresight to detect adulteration. Hence, scientists have come up with various technologies to check the same. Most traditional methods to check adulteration in food items make use of chemicals and thermal routes, which could not only fail but are complicated and difficult to use.

In this electronic era, electronic devices have influenced almost all aspects of life including medical, entertainment, safety, sports, education and defence. Electronic sensing is set to play a great role in checking adulteration of not just food items but other commodities as well.

Electronic sensing is not limited to reproducing human senses using sensors and pattern-recognition systems. The various stages of electronic sensing for recognition process are similar to human olfaction. These are performed for identification, comparison, quantification and other applications, including data storage and retrieval.

In the last few decades, electronic sensing technologies and devices have undergone significant developments from technical and commercial points of view.

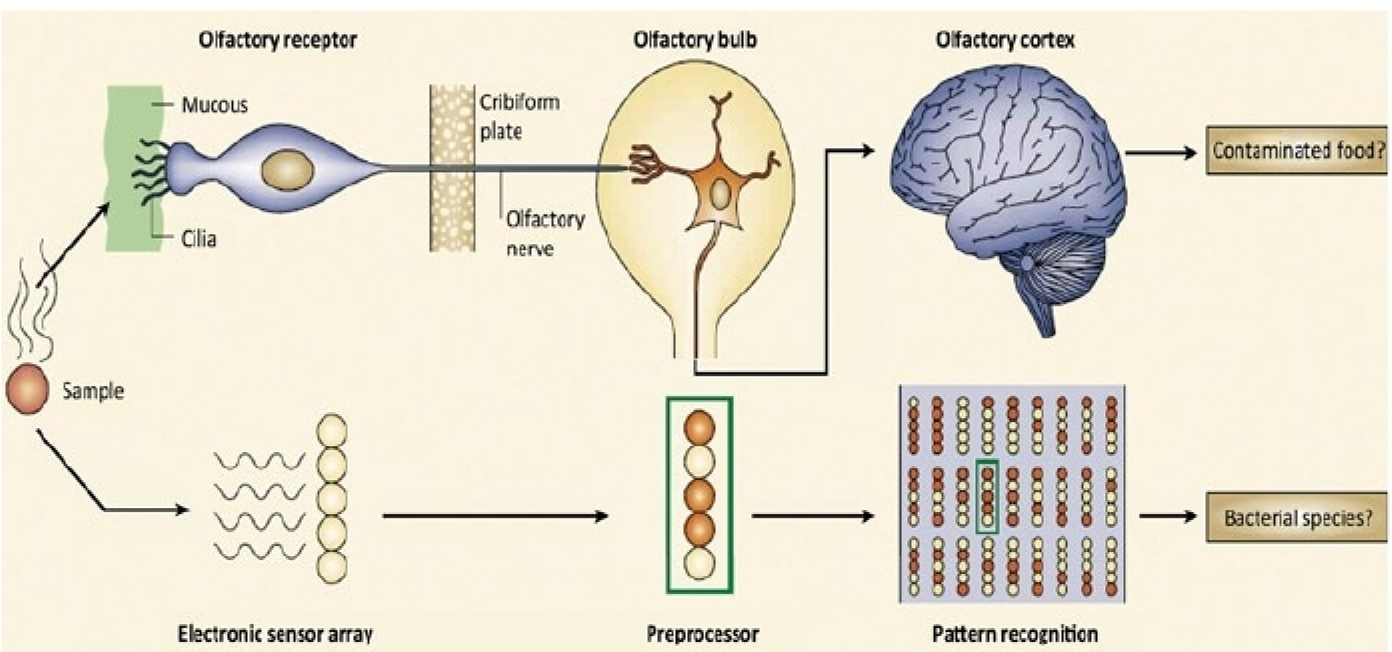


*Fig. 1: Electronic nose identifies specific components of an odour and analyses its chemical makeup to identify the same*